

Discretization Methods

Consider Fredholm integral equations of the first kind

$$\int_0^1 K(s, t) f(t) dt = g(s), \quad 0 \leq s \leq 1.$$

Replace the problem of computing the unknown function f with a “discrete” problem $Ax = b$ with $A \in \mathbb{R}^{n \times n}$ (here, square A for simplicity).

Quadrature Methods (aka Nyström methods). Compute approximations $\tilde{f}_j \approx f(t_j)$ to the solution at n selected abscissas $t_1, t_2, \dots, t_n$.

Expansion Methods. Compute an approximation of the form

$$f^{(n)}(t) = \sum_{j=1}^n \zeta_j \phi_j(t), \quad 0 \leq t \leq 1,$$

where $\phi_1(t), \dots, \phi_n(t)$ are expansion/basis functions.

The solution vector x (in $Ax = b$) consists of either $\{\tilde{f}_j\}$ or $\{\zeta_j\}$.

Quadrature Discretization

Recall the generic **quadrature rule**

$$\int_0^1 F(t) dt = \sum_{j=1}^n w_j F(t_j) + E_n ,$$

where E_n is the quadrature error, and

$$w_j = \text{weights} , \quad t_j = \text{abscissas} , \quad j = 1, \dots, n .$$

Now apply this rule *formally* to the integral with $F(t) = K(s, t) f(t)$:

$$\Psi(\mathbf{s}) = \int_0^1 K(\mathbf{s}, t) f(t) dt = \sum_{j=1}^n w_j K(\mathbf{s}, t_j) f(t_j) + E_n(\mathbf{s}) .$$

Since K is a function of $\mathbf{s}$, then so is the integral.

Quadrature Discretization + Collocation

Collocation (Wikipedia): select that solution which satisfies the given equation at a number of collocation points.

Here, we enforce the collocation requirement that Ψ equals the right-hand side g at n selected collocation points $s_1, s_2, \dots, s_n$:

$$\Psi(s_i) = \sum_{j=1}^n w_j K(s_i, t_j) f(t_j) + E_n(s_i) = g(s_i), \quad i = 1, \dots, n.$$

We can think of $g(s_i)$ as sampled or measured values of the function g .

Neglect unknown error term $E_n(s_i)$ and, consequently, replace $f(t_j)$ by $\tilde{f}_j$:

$$\sum_{j=1}^n w_j K(s_i, t_j) \tilde{f}_j = g(s_i), \quad i = 1, \dots, n.$$

Could use $m > n$ collocation points $\rightarrow$ overdetermined system.

Quadrature Method: The Matrix Form

Write out the last equation to obtain

$$\begin{pmatrix} w_1 K(s_1, t_1) & w_2 K(s_1, t_2) & \cdots & w_n K(s_1, t_n) \\ w_1 K(s_2, t_1) & w_2 K(s_2, t_2) & \cdots & w_n K(s_2, t_n) \\ \vdots & \vdots & & \vdots \\ w_1 K(s_n, t_1) & w_2 K(s_n, t_2) & \cdots & w_n K(s_n, t_n) \end{pmatrix} \begin{pmatrix} \tilde{f}_1 \\ \tilde{f}_2 \\ \vdots \\ \tilde{f}_n \end{pmatrix} = \begin{pmatrix} g(s_1) \\ g(s_2) \\ \vdots \\ g(s_n) \end{pmatrix}$$

or simply

$$Ax = b$$

where A is $n \times n$ and we have

$$\left. \begin{aligned} a_{ij} &= w_j K(s_i, t_j) \\ x_j &= \tilde{f}_j \\ b_i &= g(s_i) \end{aligned} \right\} \quad i, j = 1, \dots, n.$$

Quadrature Method: Example foxgood

The test problem foxgood in Regularization Tools uses

$$K(s, t) = (s^2 + t^2)^{\frac{1}{2}}, \quad g(s) = \frac{1}{3} \left((1 + s^2)^{\frac{3}{2}} - s^3 \right),$$

and the solution is $f(t) = t$. We use the midpoint quadrature rule

$$w_j = \frac{1}{n} \quad \text{and} \quad t_j = h \left(j - \frac{1}{2} \right), \quad j = 1, 2, \dots, n$$

and the collocation points

$$s_i = h \left(i - \frac{1}{2} \right), \quad i = 1, 2, \dots, n.$$

Here is the MATLAB code:

```
h = 1/n;
t = h*((1:n)' - 0.5);
A = h*sqrt((t.^2)*ones(1,n) + ones(n,1)*(t.^2)');
x = t;
b = ((1+t.^2).^1.5 - t.^3)/3;
```

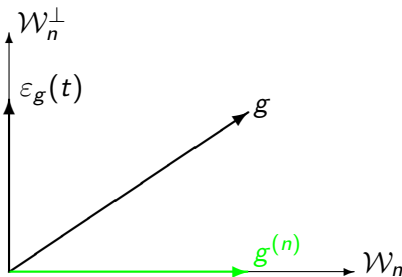
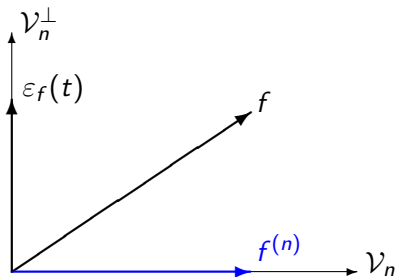
Expansion Method: the Galerkin Method I

Galerkin (Wikipedia): convert a continuous operator problem to a discrete problem by applying finite sets of basis functions.

Select two sets of expansion functions ϕ_i and ψ_j , and write

$$f(t) = f^{(n)}(t) + \varepsilon_f(t), \quad f^{(n)} \in \mathcal{V}_n = \text{span}\{\phi_1, \dots, \phi_n\},$$

$$g(s) = g^{(n)}(s) + \varepsilon_g(s), \quad g^{(n)} \in \mathcal{W}_n = \text{span}\{\psi_1, \dots, \psi_n\}.$$



Known: g and thus $g^{(n)}$. What we can compute: $f^{(n)} \in \mathcal{V}_n$.

Expansion Method: the Galerkin Method II

Write $f^{(n)}$ as the expansion

$$f^{(n)}(t) = \sum_{j=1}^n \zeta_j \phi_j(t)$$

and define the function

$$\vartheta(s) = \int_0^1 K(s, t) f^{(n)}(t) dt = \sum_{j=1}^n \zeta_j \int_0^1 K(s, t) \phi_j(t) dt .$$

If we could specifically choose the ψ_j basis functions such that

$$\psi_j(s) = \int_0^1 K(s, t) \phi_j(t) dt , \quad j = 1, \dots, n$$

then we have

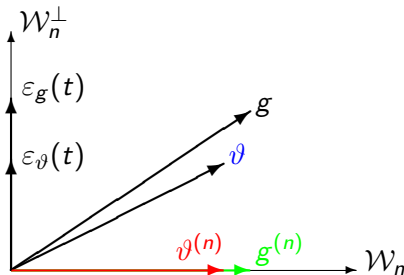
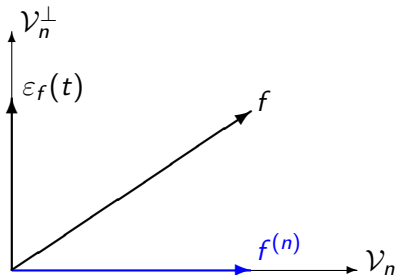
$$\vartheta(s) = \sum_{j=1}^n \zeta_j \psi_j(s) \in \mathcal{W}_n = \text{span}\{\psi_1, \dots, \psi_n\} .$$

But this is inconvenient. Hence, with “arbitrary” ψ_j the function ϑ will typically not lie in the same subspace $\mathcal{W}_n$ as $g^{(n)}$.

Expansion Method: the Galerkin Method III

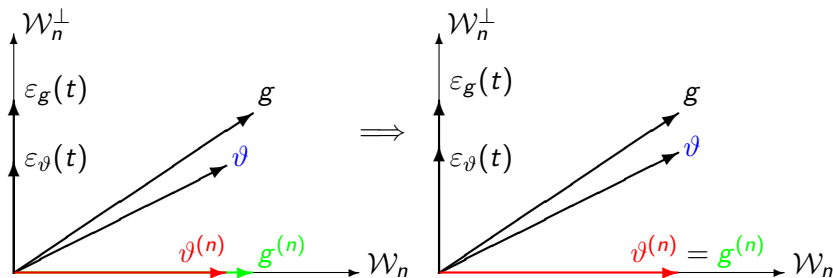
For general choices of the expansion functions we have

$$\begin{aligned}\vartheta(s) &= \int_0^1 K(s, t) f^{(n)}(t) dt = \sum_{j=1}^n \zeta_j \int_0^1 K(s, t) \phi_j(t) dt \\ &= \vartheta^{(n)}(s) + \varepsilon_{\vartheta}(s), \quad \vartheta^{(n)} \in \mathcal{W}_n = \text{span}\{\psi_1, \dots, \psi_n\}.\end{aligned}$$



Expansion Method: the Galerkin Method IV

The best we can do is to choose coefficients ζ_j such that $\vartheta^{(n)}(s) = g^{(n)}(s)$ for $s \in [0, 1]$.



This is equivalent to requiring that the residual $g(s) - \vartheta(s)$ is orthogonal to $W_n = \text{span}\{\psi_1, \dots, \psi_n\}$:

$$\langle \psi_i, g - \vartheta \rangle = 0, \quad i = 1, \dots, n$$

or

$$\langle \psi_i, g \rangle = \langle \psi_i, \vartheta \rangle, \quad i = 1, \dots, n.$$

Expansion Method: The Matrix Form

The orthogonality condition leads to the equations

$$\langle \psi_i, g \rangle = \langle \psi_i, \vartheta \rangle = \left\langle \psi_i, \int_0^1 K(s, t) f^{(n)}(t) dt \right\rangle, \quad i = 1, \dots, n.$$

Inserting the expansion for $f^{(n)}$, we obtain the $n \times n$ system

$$A x = b$$

with $x_i = \zeta_i$ and

$$a_{ij} = \int_0^1 \int_0^1 \psi_i(s) K(s, t) \phi_j(t) ds dt,$$

$$b_i = \int_0^1 \psi_i(s) g(s) ds.$$

We may need numerical quadrature to evaluate these integrals.

Expansion Method: Example deriv2

The test problem deriv2 in Regularization Tools uses

$$K(s, t) = \begin{cases} s(t-1), & s < t \\ t(s-1), & s \geq t \end{cases} \quad g(s) = \frac{1}{6}(s^3 - s),$$

and the solution is $f(t) = t$.

We use the Galerkin method with orthonormal “top hat” basis functions

$$\phi_j(t) = \begin{cases} \frac{1}{\sqrt{h}}, & (j-1)h \leq t < jh \\ 0, & \text{else} \end{cases} \quad \psi_i(s) = \text{ditto}$$

and we can evaluate the integrals analytically:

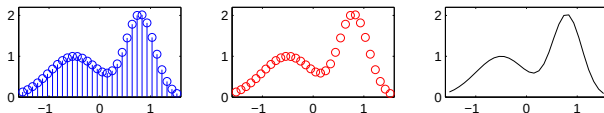
$$a_{ij} = \begin{cases} h^2 \left((i^2 - i + \frac{1}{4})h - (i - \frac{2}{3}) \right), & i = j \\ h^2 (j - \frac{1}{2}) \left((i - \frac{1}{2})h - 1 \right), & i \neq j \end{cases}$$

$$b_i = \frac{1}{6} h^{\frac{3}{2}} \left(i - \frac{1}{2} \right) \left((i-1)^2 \frac{h^2}{2} - 1 \right).$$

Final Words

Quadrature methods: Simple to use and implement (we just “sample” K and g); can avoid sampling at singularities in K ; can handle an infinite integration interval (e.g., inverse Laplace transform).

Note that we only compute an approximation to f at discrete points:



Expansion methods: Perhaps more difficult to derive and implement (single/double integrals); the expansion functions can be tailored to the application; the choice of delta functions $\psi_i(s) = \delta(s - s_i)$ represents a sampling of g .

Remember to plot the expansion – not the expansion coefficients:

